5.5 Graphic interfaces

Project: Electrical Consumption Monitor — EnergyMonitor **Record:** 3145556 — Software Analysis and Development (ADSO), SENA CIES Regional Huila **Version:** 1.0

1. Purpose and relationship with point 03

This section documents the **user interface design**: the design system that supports it, the palette, the typography, the screen structure, the reusable components, the interaction patterns and the accessibility decisions.

> **The mockups of this project are in point 03 — Software analysis**, not here. This document **does not** replace them: it describes them, classifies them and relates them to the code that implements them. When this document and a mockup seem to contradict each other, **the mockup prevails**, because it is the source of truth of the design with which the implementation was validated.

Artifact	Location	What it contributes
Mockups in HTML (web and mobile)	`03-analisis-software/mockup/layout/{web,mobile}/`	Approved visual look of each screen
Navigation maps web and mobile	`03-analisis-software/EN/navigation/`	Actual execution flow, taken from the code rather than from the mockups
Requirements and user stories	`02-especificacion-requisitos/`	Functional requirements that give rise to the screens

2. User interface design principles

#	Principle	Concrete application
1	**Clarity over complexity**	Consumption is understood with three high-level indicators (current power, today's consumption, active devices) and the detail is one level below
2	**Progressive information**	The Dashboard shows home cards; the detail opens in six tabs, never all at once
3	**One screen, one task**	Destructive actions (delete home, leave home, unlink device) are isolated in a confirmation dialog
4	**Immediate feedback**	Every on-screen change is visually confirmed: badge in the menu, tab title, toast and counter update
5	**Consistency between platforms**	Web and mobile share palette, typography, spacing and behavior; a user of both recognizes the application
6	**Real internationalization**	Four languages throughout the interface, including error messages, day names and number formats
7	**Explicit states**	Every data view declares its four states: loading, error, empty and content

3. Design system

3.1 Architecture decision

Both front-ends implement an own design system. No third-party component library is used.

Technology	Used?	Comment
CSS Modules (web)	**Yes**	Main styling technology, 40+ files
CSS variables (web)	**Yes**	Design tokens in `:root`
`StyleSheet` **+ tokens** (mobile)	**Yes**	Replication of the web system

Technology	Used?	Comment
Tailwind CSS	No	Without configuration, without dependency
shadcn/ui · Radix · Headless UI · MUI · Chakra	No	Without dependency
Icons	`lucide-react` (web) · `Ionicons` and `MaterialCommunityIcons` (mobile)	Only third-party visual library

Consequence of the decision. The project does not inherit design decisions from others: each control reflects a concrete need of this application. The cost is maintaining the system: each own component needs tests, documentation and attention to accessibility.

3.2 Components of the system

Web — 7 components in `src/design/`

Component	Variants	Function
`Button`	`primary`, `secondary`, `ghost`, `danger`; sizes `small`, `medium`, `large`	Base button of the whole application
`Input`	`default`, `otp`, `bare`	Form field with label, actionable icon and code mode
`Card`	—	Surface container with optional title
`Header`	—	Inner page header with breadcrumbs and action area
`BreadCrumb`	—	Breadcrumbs; the item with a path is a link, the current one is text
`ActionMenu`	4 positions	Dropdown menu of actions
`Badge`	`filled`, `outline`	Status badge

Mobile — 18 components in `src/components/`

Component	Variants	Function
`Button`	4 variants, 3 sizes	Base button (`Pressable`)
`Card`	`default`, `soft`, `selected`; 4 padding levels	Surface; it becomes pressable if it receives `onPress`
`Input`	`default`, `otp`	Field with internal focus state
`Modal`	`default`, `danger`	Centered dialog with appearance animation
`Alert`	`success`, `warning`, `danger`, `info`	Banner with icon, title and message
`Header` / `Breadcrumb`	—	Header and breadcrumbs
`ActionMenu`	—	Button with menu anchored by position measurement
`OtpInput`	Configurable length	Single-digit code boxes
`Switch`	—	Switch with the theme colors
`Chart`	`area`, `bar`, `pie`	Charts wrapper
`Loader`	`small`, `large`	Loading indicator with text
`EmptyState`	`full`, `compact`	Empty state with image or icon
`ListItem`	—	Reusable list row
`LanguageSwitcher`	`dark`, `light`	Language selector with flag
`Sidebar`, `BottomTabBar`, `AuthGate`	—	Layout and navigation components

> **Verified parity.** Seven of the web system components have a direct equivalent on mobile, with the same variants. It is evidence that this is a single design, not two.

3.3 Color palette

Figure 8. *Brand palette and semantic colors of the design system.*

Brand and semantic colors

Token	Value	Use
`--color-primary`	`#0078D7`	Primary button, links, focus indicator
`--color-primary-hover`	`#3399FF`	Pressed state, focused field border
`--color-primary-dark`	`#0A2540`	Gradient end, limit bar background
`--color-primary-soft`	`#E8F3FF`	Subtle background of active tabs
`--color-secondary`	`#32CD32`	Accent and success green
`--color-success`	`#2ECC71`	Confirmation
`--color-warning`	`#FFD700`	Warning
`--color-danger`	`#E63946`	Destructive action
`--color-info`	`#17A2B8`	Information

Surfaces and text

Token	Value
`--color-background`	`#F8F9FA`
`--color-background-left`	`#3F6BAE`
`--color-surface`	`#FFFFFF`
`--color-text-primary`	`#212529`
`--color-text-secondary`	`#6C757D`
`--color-border`	`#E5E7EB`

State pairs — each state has a soft background and a text color with sufficient contrast

State	Background	Text
Error / critical	`#FCEBEB`	`#A32D2D`
Success / online	`#EAF3DE`	`#3B6D11`
Warning	`#FFF4D6`	`#854F0B`
Information	`#D1ECF1`	`#0C5460`

> **Mobile replicates the 32 color tokens** in its `ThemeColors` contract, with the same hexadecimal values. The parity is documented in the code itself with comments that refer to the source file on the web side.

3.4 The four themes

Requirement RF-CLI-06 (color configuration) is implemented with **four themes**, two light and two dark. The preference is stored on the device (`energymonitor_theme`) and in the database (`user_configuration.color_theme`).

Theme	Mode	Sidebar	Primary	Character
EnergyMonitor *(default)*	Light	`#3F6BAE`	`#0078D7`	Corporate, sober
Eco Hogar	Light	`#2E7D32`	`#32CD32`	Natural, oriented to savings
EnergyMonitor Dark	Dark	`#0F172A`	`#1E293B`	High-contrast dark

Theme	Mode	Sidebar	Primary	Character
Eco Dark	Dark	`#102A1A`	`#183321`	Dark with a green hue

True dark mode, not inverted. In the dark themes each token is reassigned: no inversion filter is applied over the light theme. The soft colors go from solid hexadecimals to translucent values (`rgba(…, 0.15)`), and the danger color is lightened to `#FCA5A5` to maintain contrast on dark backgrounds.

Preview in the interface. The configuration screen draws a thumbnail of each theme —sidebar, progress bar and color dots— built with the same values that will be applied. The user chooses by looking, not by reading.

3.5 Chart colors

Charts use a fixed categorical palette, independent of the theme, with a **variant for light and dark mode**:

Appliance	Light	Dark
Refrigerator	`#2563EB`	`#60A5FA`
Washing machine	`#7C3AED`	`#A78BFA`
Television	`#DB2777`	`#F472B6`
Microwave	`#EA580C`	`#FB923C`
Air conditioner	`#0D9488`	`#2DD4BF`
Computer equipment	`#CA8A04`	`#FACC15`
Water heater	`#DC2626`	`#F87171`
Lighting	`#65A30D`	`#A3E635`
Other	`#64748B`	`#94A3B8`

> **Why this palette.** The colors are not chosen at random: each one maintains a perceptible contrast relationship with the others, so that two adjacent segments of a pie chart are never confused. In addition, the same appliance always has the same color, which allows recognizing it from one screen to another without consulting the legend.

3.6 Typography

Families

Platform	Primary font	Secondary font
Web	`Inter`	`Poppins`
Mobile	`System` (operating system font)	`System`

> **Documented technical debt.** The web declares `Inter` and `Poppins` in its CSS variables, but does not load either of them through `@font-face` nor from a CDN. The browser falls back to the generic system font. To fix it, it is enough to add the `@font-face` or the corresponding link in `index.html`.

Typographic scale (11 sizes, identical on both platforms)

Token	Value	Typical use
`size11`	11 px	Chart text and legend labels
`xs`	12 px	Auxiliary text
`size13`	13 px	Small labels
`sm`	14 px	Control text
`md`	16 px	**Base body text**
`size18`	18 px	Highlighted text

Token	Value	Typical use
`lg`	20 px	Subtitles
`size22`	22 px	Card titles
`xl`	24 px	Section titles
`xxl`	32 px	Screen titles
`xxxl`	60 px	Cover messages

Weights: `normal` (400), `medium` (500), `bold` (600).

> The scale is deliberately irregular: the "round" names (`xs`, `sm`, `md`, `lg`, `xl`, `xxl`) replicate the web scale to preserve visual parity, and the "exact" ones (`size11`, `size13`, `size18`, `size22`) cover sizes that exist in the design but did not fit the scale.

3.7 Spacing and radii

Spacing — 4-point scale

Token	Value
`xs`	4 px
`sm`	8 px
`md`	16 px
`lg`	24 px
`xl`	32 px

Border radii — `sm` 6 px · `md` 10 px · `lg` 16 px · `999px` for pills and avatars.

Shadows

Token	Web value	Android elevation
`shadow-sm`	`0 1px 2px rgba(0,0,0,0.05)`	1
`shadow-md`	`0 4px 6px rgba(0,0,0,0.1)`	4
`shadow-lg`	`0 10px 15px rgba(0,0,0,0.25)`	10

> **Mobile detail.** Each shadow is a pair: the `boxShadow` string for iOS and web, and the `elevation` value for Android. Defining only one of the two produces invisible shadows on one of the platforms.

3.8 Breakpoints (responsive)

Token	Value	Device
`--bp-mobile`	480 px	Small mobile
`--bp-tablet`	768 px	Tablet and mobile
`--bp-laptop`	1024 px	Laptop

Most relevant responsive transformations

Component	Desktop	≤ 1024 px	≤ 768 px
Sidebar	Fixed, 240 px, collapsible to 80 px	Same	Off-screen sliding layer, 85 % of the width with a maximum of 300 px
Top navigation	Complete with links	Reduced typography	Expandable hamburger menu
Home detail tabs	Horizontal with text	Fixed bar at the bottom	Icons only, with bottom bar

Component	Desktop	≤ 1024 px	≤ 768 px
Breadcrumbs	All levels	All levels	Only first and last level, with horizontal scrolling
Dialogs	Centered	Centered	**Bottom sheet** (*bottom sheet*) attached to the bottom edge
Brand panel of the login	50 % left	Padding adjustments	Hidden; the form takes the full width
Progress bars	Full width	Full width	Taller bars to make touching easier

> **Mobile application strategy.** The app is locked to portrait orientation (`orientation: "portrait"` in `app.json`). There is no landscape design to solve.

4. Screen structure — Web application

Figure 9. *Navigation map of the web application.*

4.1 Route table

Route	Screen	Access	Layout
`/`	Redirect to `/home`	Public	—
`/home`	Welcome page	Public	Navigation + cover + benefits
`/about`	About us	Public	Navigation + content
`/support`	Support and frequently asked questions	Public	Navigation + question search
`/login`	Sign in	Public	Brand panel + form
`/register`	Sign up	Public	Brand panel + form
`/VerifyAccount`	Email verification	Public	Brand panel + 6-digit code
`/recover-password`	Password recovery (step 1)	Public	Brand panel + form
`/verifyRecoverPassword`	Verification and new password (step 2)	Public	Brand panel + form
`/dashboard`	**Home panel**	**Protected**	Sidebar + home cards
`/homes/:homeId`	**Home detail**	**Protected**	Sidebar + six tabs
`/favorites`	Favorites	**Protected**	Sidebar + cards
`/notifications`	Notifications (ADMIN: administration notices)	**Protected**	Sidebar + four tabs and a home filter
`/settings`	Settings	**Protected**	Sidebar + four sections
`/account`	My account	**Protected**	Own screen inside the protected zone
`/admin/users`	**Administration: platform users**	**Protected +** `USER.READ` **permission**	Administrator sidebar + search + paginated table
`*`	Not found page	Public	Message and back link

4.2 Protection levels

Routing applies **five guards** with distinct responsibilities:

Guard	What it verifies	If it fails
`ProtectedRoute`	That a session with an access token exists	Redirects to `/login`
`HomeRoute`	That the home in the URL exists and is accessible	Shows "home not found" with a back button

Guard	What it verifies	If it fails
`GuestRoute`	That there is no session when opening `/login` or `/register`	Goes to the dashboard
`UserOnlyRoute`	That the account is not an administrator (home screens)	Goes to `/admin/users`
`RequirePermission`	That the account has the given global permission (`USER.READ` on `/admin/users`)	Goes to `/dashboard`; the API answers 403 anyway

Figure 10. *Decision flow for access to a protected route.*

Design decision — outlet context. `HomeRoute` delivers the home and the owner flag (`isOwner`) to the child screen through the router's outlet context. This avoids passing the home as a property through three levels of components and, above all, avoids having each screen query it again.

4.3 Detail of the main screens

Welcome page

Aspect	Detail
Elements	Top navigation with Home, Benefits, How it works, About us and Support; language selector; theme button; "Sign in" and "Sign up" buttons; welcome message; benefits section in cards; "How it works" section
Objective	Explain what the system does in less than a minute and offer direct access
Design	Composition in two blocks: presentation and call to action

Home panel (Dashboard)

Aspect	Detail
Elements	Sidebar with home, favorites, notifications and settings; avatar indicator at the bottom; cards of the user's homes; "+" quick access button
States	**Loading** — indicator with text; **Error** — translated message and "Retry" button, with a distinction between "no connection" and "server error"; **Empty** — invitation to create the first home with image and button; **Content** — grid of cards
Home card	Name, identifying header color, description, favorite button (heart) and access to the detail
Actions	Create home (name, type, address, description) · Join a home (8-character code, with counter and automatic uppercase conversion)

Home detail — Consumption tab (default tab)

Aspect	Detail
Indicators	Three cards: **current power** (kW and risk level), **today's consumption** (kWh against the limit), **active devices** (active over total)
Charts	**Area** — average hourly active power over the last 24 hours · **Pie** — percentage distribution of consumption by appliance type · **Horizontal bars** — current consumption per device
Limit bars	Two progress bars: progress of the daily limit and of the monthly limit
States	Chart without data → empty component with icon, **never** a chart at zero

Home detail — History tab

Aspect	Detail
Selectors	Period: day, week, month or year. Within the month: specific week or "show all"
Chart	Bars with the distribution of consumption by device over the period

Aspect	Detail
Table	Device ranking with a percentage bar per row
Cards	Total consumption of the period, average and the appliance that consumes the most
Behavior	If there are more than 8 points, the chart scrolls horizontally instead of being compressed

Users, Devices, Thresholds and Home tabs

Tab	Owner	Member
Users	List of members with role; can expel a member and transfer ownership to them ("Make … the responsible user", with confirmation)	List of members; read only
Devices	List with status and consumption; link, edit and unlink	List with status and consumption; read only
Thresholds	Default values switch and daily and monthly limit fields; "Save changes" button	Viewing of the limits; read-only fields
Home	Home data, owner data, access code to share and "Delete home" button (confirmed by typing the home name)	Home and owner data; "Leave home" button

> **Interface rule.** The quick access button and the sidebar elements are shown from the start: the user discovers the available functions. However, **actions that the role does not allow are hidden** (for example, the "Delete home" button does not appear for a member).

Notifications screen

Aspect	Detail
Tabs	All · Alerts · Recommendations · Activity, each with a pending counter, and a filter by home or "My account"
Bulk action	"Mark all as read" and "Delete resolved or read" (with confirmation)
Card	Icon according to the type · title · description · related home and device · date and time · status tag (New, Active or Resolved) · action (Resolve, Mark as read or delete)
Iconography	Connectivity → wifi off · Device → meter · Limit → gauge · Critical → lightning bolt · Warning → triangle · Recommendation → light bulb

Settings screen

Section	Content
Appearance	Grid of the four themes with preview thumbnail
Language	Selector for Spanish, English, French or Portuguese, with flag
Notifications	Email, push and combined switches; indication of subscribed browsers
Help	Documentation, contact and updates with its own brand icon (marked "Coming soon")

An administrator account does not see the **Notifications** section of the settings.

Administration screens (web only, ADMIN role)

Aspect	Detail
Sidebar	Administration · Notifications · Settings (no Home, Homes or Favorites)
Platform users (`/admin/users`)	Breadcrumb Home › Administration; title and subtitle; search by name or email; table User (with the "You" mark) · Status · Global roles · Registered · Last access; pages of 20

Aspect	Detail
Row actions	Active / Inactive / Blocked status selector (`USER.MANAGE` permission) and shield button to grant or revoke ADMIN (`ROLE.MANAGE`); hidden on one's own row and without the permission
Confirmations	`ConfirmModal` when leaving Active and always when granting or revoking the role
Administration notifications	List of notices (account registered, status changed, role granted or revoked) with New/Read, mark as read, mark all and delete
Mockups	Do not exist yet (see `03-analisis-software/EN/navigation/README.md`, section 6.1)

Figure 10b. *"Platform users" screen with the status change confirmation* — image pending.

5. Screen structure — Mobile application

Figure 11. *Navigation map of the mobile application.*

5.1 Route table

Route	Screen	Access
`/`	Sign in	Public
`/register`	Sign up	Public
`/recover-password`	Password recovery	Public
`/verify-account`	Email verification	Public
`/verify-recover-password`	Recovery verification	Public
`/new-password`	New password	Public
`/dashboard`	**Home panel**	**Protected**
`/homes`	Redirects to `/dashboard`	**Protected**
`/home/[id]`	**Home detail**	**Protected**
`/favorites`	Favorites	**Protected**
`/notifications`	Notifications	**Protected**
`/settings`	Settings	**Protected**
`/account`	Account	**Protected**

13 routes · 12 functional screens. The app has no landing page and no administration screens; the registration opens the terms and the privacy policy in a window, and Settings and My account have a page title.

5.2 Navigation architecture

Element	Implementation	Design decision
Stack	A single `Stack` of Expo Router, without header	Simple and predictable structure
Sidebar	Own component with `Animated` and 300 ms transition	Allows an accordion with the home list synchronized with the global context
Tab bar	Own component in the home detail	The "back" button exits the home and not the section
Context menu	`ActionMenu` with position measurement (`measureInWindow`)	The dropdown is anchored correctly even if the layout changes

Figure 12. *Mobile navigation structure: single stack, own sidebar and own tab bar.*

Navigation guard. `AuthGate` wraps the complete stack. It defines the set of public routes and, on detecting that the session is not valid, redirects to the initial screen. While it restores the session from storage it renders nothing, to avoid a flicker of the loading screen.

5.3 Home detail — bottom bar of six sections

Section	Icon	Content
Consumption	Pulse	Three indicators, 24 h area chart, distribution pie chart and bars per device, daily and monthly limit bars
History	Clock	Period selector, bar chart and ranking table with percentage bars
Users	People	List of members with role; the owner can expel
Devices	Wireless	List with connection status and consumption; the owner can link, edit and unlink
Thresholds	Sliders	Defaults switch and limit fields with period selector
Home	House	Home data, owner data and access code

5.4 Adaptation of screens to different sizes

Technique	Application
Safe areas	The 13 screens apply `paddingTop: max(insets.top, spacing.md)` and respect the bottom margin of the device
Derived widths	Dialogs set a maximum width of 440 px, avoiding giant dialogs on tablet
Sidebar	Width of `min(85 % of the screen, 300 px)`
Derived heights	The long lists of the dialogs occupy at most 55 %, 62 % or 45 % of the height depending on their content
Chart scrolling	If the series has more than 8 points, the chart scrolls horizontally with a width of `points × 56 px`
Keyboard	Nine screens use the container that avoids the keyboard, with different behavior on iOS and Android
Fluid grids	The theme, language and home selectors use wrapping with a base width of 45 %

6. Interaction patterns

6.1 Forms and validation

Aspect	Web	Mobile
Engine	`react-hook-form` + `zod`	`react-hook-form` + `zod`
Schemas	6 authentication and account schemas	10 schemas, including home and thresholds
Messages	Function that receives the translator: the message is written in the active language	Same, through a central helper
Password	Visibility eyes, complexity rules, matching confirmation	Same
6-digit code	6 boxes with automatic advance and backspace	`OtpInput` component with the same behavior

> **Two-layer validation.** The Zod schema acts on the client to give an immediate response; the backend validates everything again with Bean Validation. Client validation never replaces server validation: it is a usability aid, not an integrity guarantee.

6.2 States of a data view

Every screen that consumes information declares the same four states. This pattern is the backbone of visual consistency.

Figure 13. *State machine of a data view.*

State	Presentation	Action offered
Loading	Indicator with text and `status` role	None; one waits

State	Presentation	Action offered
Network error	Translated message + wifi off icon	"Retry"
Server error	Message translated by status code + alert icon	"Retry"
Empty	Image or icon + explanatory title + primary action	"Create home", "Link device"
Content	The data	Those of the screen itself

> **Hard rule.** The error message the user sees is **never** the one returned by the server. The server responds with an HTTP code and the client translates it through a catalog that combines the operation performed with the code received. This prevents leaking internal details and guarantees that all messages are translated.

6.3 Modal dialogs

Type	Use	Behavior
Simple confirmation	Reversible or moderate actions	Title, message, "Cancel" and "Accept"
Destructive confirmation	Delete home, transfer ownership, leave home, expel member, unlink device, delete account	Same design, but with `alertdialog` role and danger color
Form	Create home, join home, link device, edit profile, change password	Fields with inline validation
Step-by-step wizard	Link a device	Five steps with progress indicators and signal bars

Common behavior

Rule	Detail
Close	Close button, Cancel or tap on the darkened background
Focus	On opening, the focus stays inside the dialog
Web — bottom sheet	On small screens the dialog becomes a sheet attached to the bottom edge
Mobile — keyboard	The keyboard does not cover the fields: the dialog scrolls
Double confirmation	No destructive action is executed without an explicit confirmation

6.4 Multichannel notification system

A single event is communicated through several channels, according to the context of the user.

Figure 14. *Multichannel flow of an alert: from the measurement to the notice on the device.*

Channel	When	Presentation
Badge in the menu	Always	Counter in the sidebar entry, with a visual cap at `99+`
Tab title	In the browser	The title becomes `(3) EnergyMonitor`
Toast	On screen	Panel that slides in, maximum 3 at a time, 8 seconds each
Browser notification	With the application open or closed	System notification, on click it focuses the application
Email	According to the user preference	Message sent via SMTP
Audio and vibration	Mobile	Reserved for the version with native notifications

System rules

Rule	Justification
Each type has its icon and its severity color	Distinction above color
The left border indicates the severity	Red for critical, amber for warning, blue for informational
Already seen notices are not repeated on reload	The identifiers already shown are stored

Rule	Justification
On tapping a notice it navigates to the inbox	The notice always leads to its origin
A network failure does not empty the inbox	The user keeps what they already had

6.5 Charts

Aspect	Web	Mobile
Library	`recharts`	`react-native-chart-kit` over `react-native-svg`
Area	24-hour curve with gradient	24-hour curve with gradient shadow
Pie	Donut with free center and separation between segments	Sector with legend
Bars	Horizontal per device	Vertical per device
Limit bars	Indicator built with CSS	Indicator built with native views
Colors	Categorical palette with a variant per theme	The same palette
Scale	Automatic	Automatic
Interaction	Tooltips with the system style	Component tooltips
Empty	`EmptyChart` component with icon and explanation	"No data" message — **never** a chart at zero

> **Design decision — "no data" is not "zero".** In both front-ends the distinction between "there are no measurements" and "the consumption was zero" is preserved. A chart showing a line at zero asserts something the system does not know; an empty chart with an explanation is honest.

6.6 Real-time data

Requirement RF-MED-01 (automatic recording of consumption) is reflected in the interface through polling at three distinct moments.

Data	Frequency	Condition
Devices and their status	5 seconds	Only with the application in the foreground (mobile)
Consumption summary	30 seconds	Only with the application in the foreground (mobile)
Alerts and recommendations of all homes	30 seconds	Only with the application in the foreground (mobile)

Connection states derived from the absence of data

State	Criterion	Presentation
Online	Recent data	Green indicator + signal strength in percentage
Verifying	More than 75 s without data	Amber indicator "checking"
Disconnected	More than 150 s without data	Gray indicator + "Disconnected"

> **Design decision.** An intermediate state of "verifying" avoids the false positive: if the device has not reported in 75 seconds, it is not declared disconnected yet. Without this state, a brief network outage would appear as a device disconnection.

6.7 Navigation and feedback

Element	Behavior
Back	On mobile, the back button exits the home detail toward the panel; inside the web application, browser routes work normally
Favorites	Heart button on each card; the applied value is the one returned by the server, not a local toggle, so that two quick taps do not leave the heart reversed

Element	Behavior
Language	The change applies immediately throughout the application and is persisted
Theme	The change applies immediately without reloading the page
Sign out	Available from the profile menu, on both platforms

7. Accessibility

Requirement RNF-USA-01: The interface must be clear, adaptable to different screen sizes and coherent, with immediate feedback. No formal WCAG conformance level is declared.

7.1 What is implemented

Category	Implementation
Roles and regions	Dialogs with `dialog` role and `aria-modal`; destructive confirmations with `alertdialog` role; switches with `switch` role; radio groups with `radiogroup` role; loading states with `status` role; errors with `alert` role
Labels	All icon-only buttons have a label; decorative icons are marked to be skipped by the screen reader
States	Switches, tabs and selected cards expose their state; accordions expose whether they are expanded
Keyboard navigation	Dialogs close with `Escape`; the numeric code advances and goes back automatically; cards and pressable fields respond to `Enter`
Visible focus	2 px outline with the primary color on buttons, menus and fields
Live regions	The toast stack uses a `polite` region; the device status is announced
Contrast	Each state has a background-text pair designed to meet contrast; the dark themes readjust the danger color to maintain it
Touch size	Controls with an enlarged touch zone in dense lists; on mobile the small touch targets were enlarged
Mobile (October 2026)	Screen reader labels on fields, icon buttons and code digits (which accept pasting); roles and states on the controls; screen titles marked as headings; side menu hidden from the reader when closed; icons instead of emojis
Document language	The `lang` attribute of the document changes with the chosen language
Regional format	Dates and numbers are formatted according to the language and time zone of the browser

7.2 Identified gaps

#	Gap	Impact	Recommendation
1	No close with `Escape` in most dialogs (only in one of the web ones)	A keyboard user has to look for the close button	Implement close with `Escape` in the base dialog component
2	No focus retention within dialogs	The focus can leave the dialog with `Tab`	Implement a focus trap and restore the focus on close
3	Tabs do not use the `tablist`, `tab` and `tabpanel` roles	A screen reader does not announce how many sections there are nor which one is active	Adopt the accessible tabs pattern
4	The graphs do not have an alternative textual description	The information of the charts is not accessible	Add a description and refer to the data table, which already exists in the history
5	Elements with `role="button"` over `div` do not respond to the spacebar	An expected keyboard action is skipped	Use the native `button` element
6	No skip link to the main content	A keyboard user has to traverse the whole navigation on each screen	Add a skip link

7.3 Accessibility-oriented tests

The test suite of the web front-end uses role queries (`getByRole`) systematically, which constitutes an involuntary accessibility check: **if a control loses its role, the test fails**. It is a cheap and effective practice worth keeping.

8. Internationalization

Aspect	Detail
Languages	Spanish (default), English, French and Portuguese
Namespaces	31 on the web · 25 on mobile
Total translation files	124 on the web · 100 on mobile
What is translated	All visible text: labels, error messages, field help, day names, units and system messages
What is not translated	The product name and the password placeholder
Regional format	Dates and numbers with `Intl` according to the active language; the time zone of the browser is sent to the backend to group the measurements by local hour
Initial language	Stored preference → device language → Spanish
Anti-flicker	On mobile, the language is resolved **before** mounting the provider, avoiding the flash of the interface in Spanish
Validation messages	The messages of the validation schemas are resolved with the active language, they are never hardcoded

Figure 15. *Languages available in the selector, with flag and language name.*

9. Consistency between platforms

9.1 Equivalence map

Web	Mobile	Equivalence
Fixed or collapsible sidebar	Sliding sidebar	Same destinations, same position of the elements
Horizontal top tab bar	Bottom bar with icons	Same six sections of the home detail
50/50 brand panel	Branded header with gradient	Same purpose: product identity versus the form
Centered dialog	Centered dialog	Same components and variants
Dialog as bottom sheet on mobile	Native centered dialog	Coherent adaptation with the platform
Top navigation with hamburger	No equivalent: the sidebar fulfils the function	Own decision of each platform
Browser notification (Web Push)	Native push notification (Expo)	Same email and push preferences
Browser Web Bluetooth	Native Bluetooth (BLE)	Same linking wizard
Administration screens	No equivalent	Decision: administration is done only on the web

9.2 Verified common elements

Element	Match
32 color tokens	Identical hexadecimal values
4 theme palettes	Same identifiers, same bar and surface colors
11 typographic sizes	Identical, including the "exact" values
Spacing scale	Identical (4, 8, 16, 24, 32)
Chart palette	Identical, with the same light and dark variant

Element	Match
Seven system components	Same variants in `Button`, `Card`, `Input`, `Header`, `Breadcrumb`, `ActionMenu`, `Alert`
Validation rules	Same set of password and email rules
Consumed endpoints	The same set of API routes
Error model	Same structure: status code and translate the message

> **Recommendation for the next iteration.** Parity is maintained by manual replication of values, with comments that refer to the source file. This works, but it is fragile: a change in the web palette may not reach mobile. The solution would be a shared package with the tokens, which both front-ends import.

10. Interface gaps and pending work

#	Gap	Platform	Impact	Pending work
1	Solved: device registration via native Bluetooth	Mobile	—	Implemented with `react-native-ble-plx`
2	Solved: copying the access code	Mobile	—	Implemented with `expo-clipboard`
3	Solved: native notifications	Mobile	—	Push with `expo-notifications`
4	Solved: Google and legal links	Mobile	—	Native Google sign-in and legal window at registration
5	Solved: fallback route	Web	—	`*` route with `NotFound.jsx`
6	Solved: lazy loading of routes	Web	—	Screens with `lazy` and `Suspense`
6b	No administration screens	Mobile	An administrator must use the web	Scope decision; keep it documented
7	No lazy loading of images	Web	All images are downloaded at startup	Add lazy loading
8	No automated test	Mobile	Verification is manual	Incorporate a testing framework
9	Unused components	Both	`Badge` (web), `ListItem` and `Breadcrumb` (mobile)	Integrate or remove them
10	Orphan routes	Both	Files that do not correspond to any route	Remove them from the code

11. Traceability with the requirements

Requirement (requirements specification)	Screen or design element
RF-SEC-05 Social network login	"Continue with Google" button in sign up and sign in
RF-SEC-08 Information update	Account screen: first name, last name and photograph
RF-SEC-06 Recover password	Two-step recovery screens
RF-SEC-11 Delete account	Deletion dialog inside the account screen
RF-HOM-01 Register a home	Home creation form
RF-HOM-05 Permission management	Difference of actions between owner and member
RF-HOM-05 Responsible user	Home tab: owner data and access code
RF-HOM-07 Favorite homes	Heart button on the cards + favorites screen
RF-HOM-08 Threshold configuration	Thresholds tab with defaults switch
RF-CLI-06 Color configuration	Grid of four themes with preview
RF-CLI-05 Language configuration	Language selector with flag
RF-NOT-03 Alert configuration	Notification switches in settings
RF-MED-08 Selection of date ranges	Period selector in the history tab

Requirement (requirements specification)	Screen or design element
RF-MED-08 Consumption comparison	Ranking table with percentage bar
RF-MED-08 Trend charts	Area, pie and bar charts
RF-ALE-05 Alert display	Notifications screen with three tabs
RF-REC-08 Recommendation display	Recommendations tab in notifications
RF-DEV-08 Device linking	Step-by-step linking wizard
RF-MED-04 to RF-MED-06 Connection status	Indicator per device with signal and derived status
RNF-USA-01 Usability and accessibility	Design system, four themes, four languages, declarative states

12. Related artifacts

Artifact	Location
Mockups	`03-analisis-software/mockup/layout/{web,mobile}/` — source of truth of the visual design
Navigation maps	`03-analisis-software/EN/navigation/`
Presentation architectural model	`05.1-architectural-model.md`, section 6
Interface design patterns	`05.4-design-patterns.md`, section 7
User manual (web and mobile, single document)	`12-manual-usuario/EN/MD/User_Manual_EnergyMonitor_EN.md`